

Robustel R1520 IPsec VPN

Guide

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Purpose

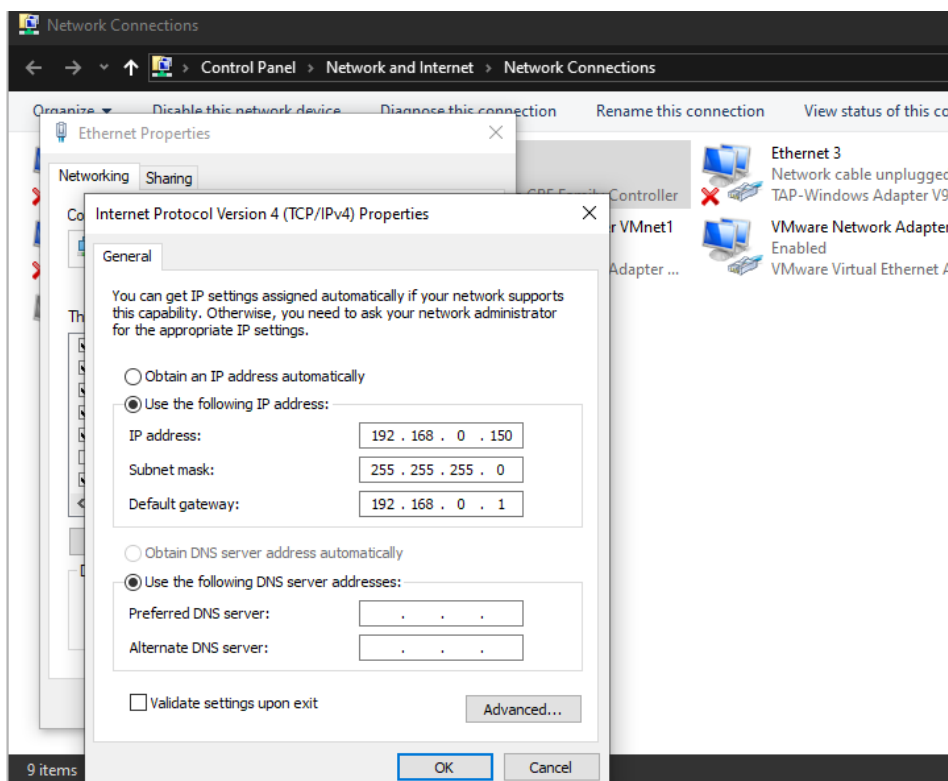
The purpose of this guide is to provide a clear, step-by-step procedure for implementing and configuring IPsec VPN on the Robustel R1520 router. This guide also covers adding a Moxa Iologik device as an end device on the PLC-side network of the Robustel, ensuring secure remote connectivity.

Topology



Configure the PC Network Settings (SCADA)

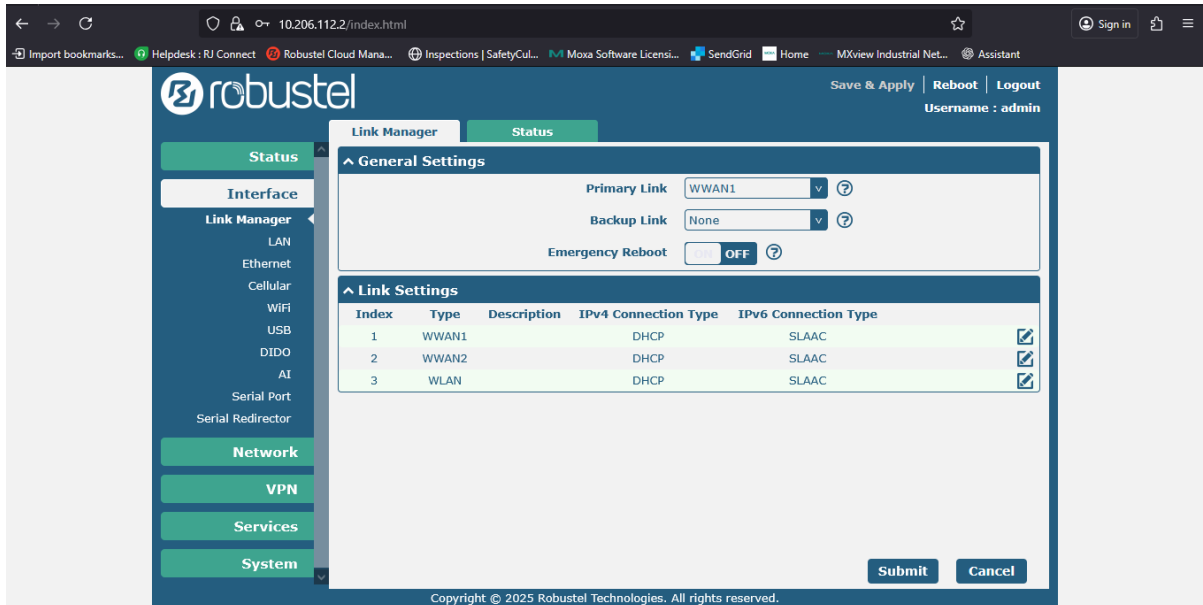
- Set the PC's IP address to be in the **same subnet range as the Robustel router** on the SCADA-side network.



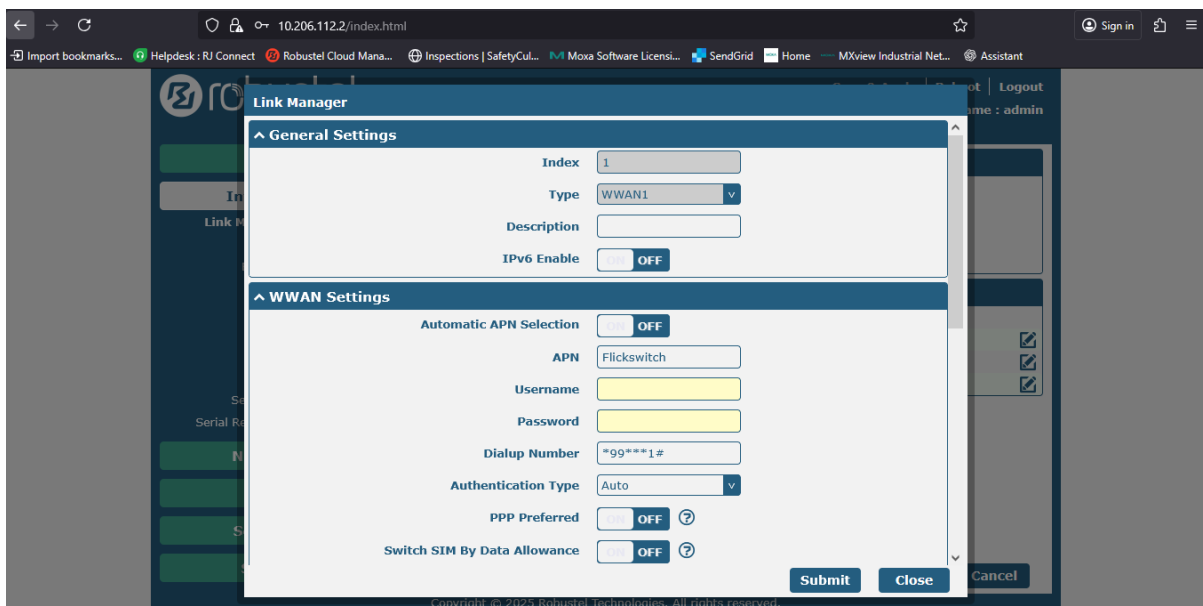
Configuration (Robustel SCADA side)

Configuring WWAN settings

- Configure **WWAN1** with the required SIM card and network settings.
- Set **WWAN1** as the **primary link** for internet connectivity.
- (Optional) If a second SIM card is available, configure **WWAN2** as the **backup link** to provide automatic failover if WWAN1 goes down.

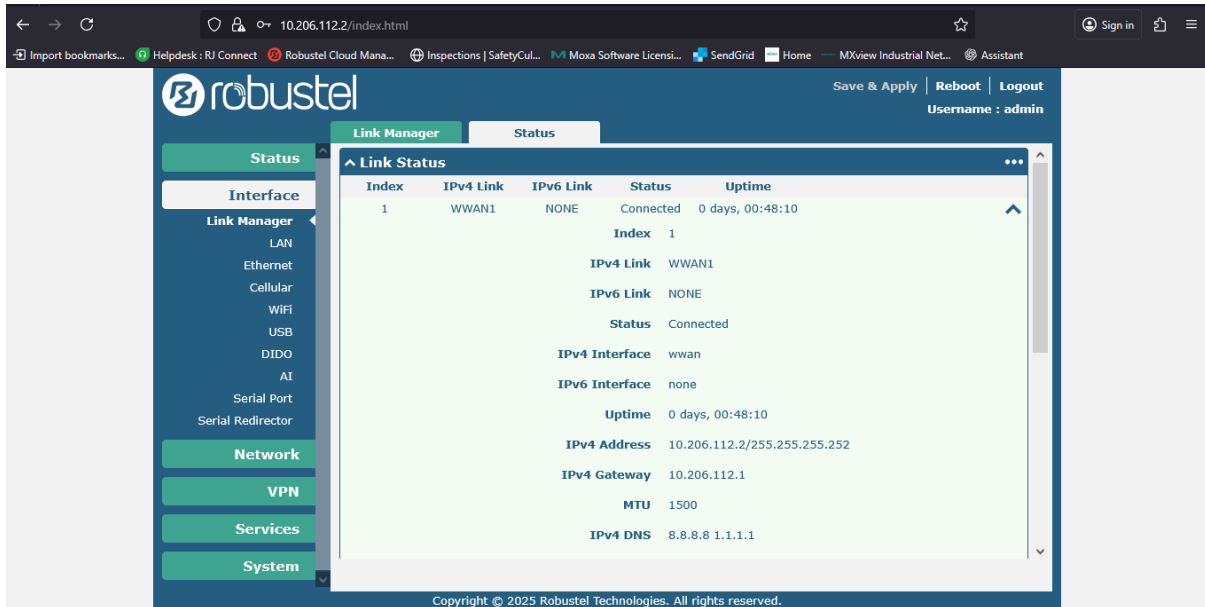


- Configure the **WWAN interface** (WWAN1 or WWAN2) according to your SIM card's required settings (APN, authentication, etc.) or your preferred network configuration.
- Ensure the interface successfully registers on the cellular network before proceeding.



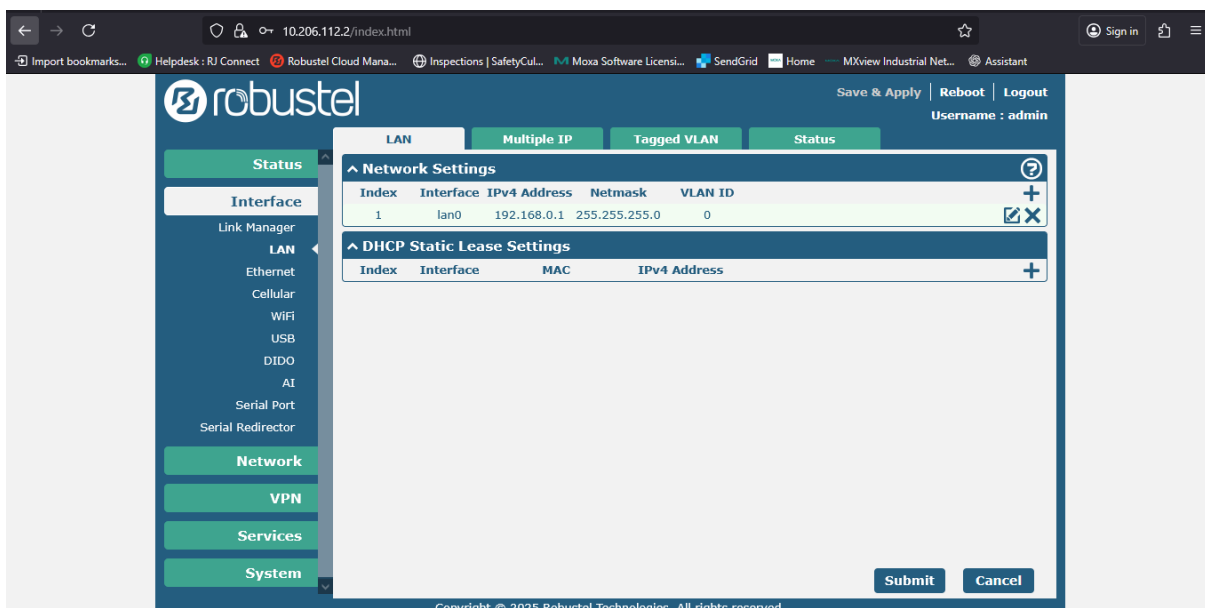
Verify SIM Card Connectivity

- Check the **WWAN status** and wait for the SIM card to register on the network.
- Confirm the status shows **Connected** before proceeding.
- Once connected, log into the **Robustel router** using the SIM card's **static public IP address**.
- **Tip:** Use *https://<SIM_IP>* in your browser to access the web interface securely.



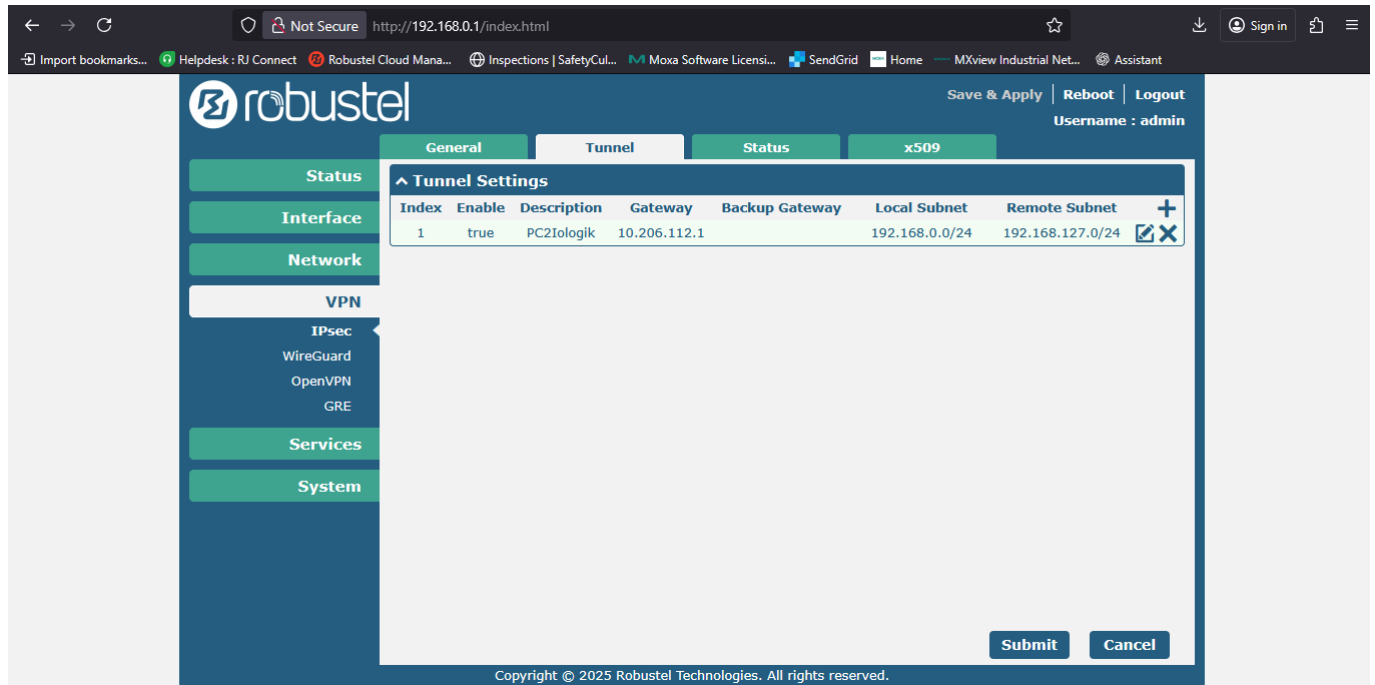
Verify LAN Settings

- Navigate to **LAN Settings** in the Robustel web interface.
- Ensure the **LAN IP address** of the router is **192.168.0.1**.
- This IP should match the **gateway address** configured on your PC to allow proper communication.



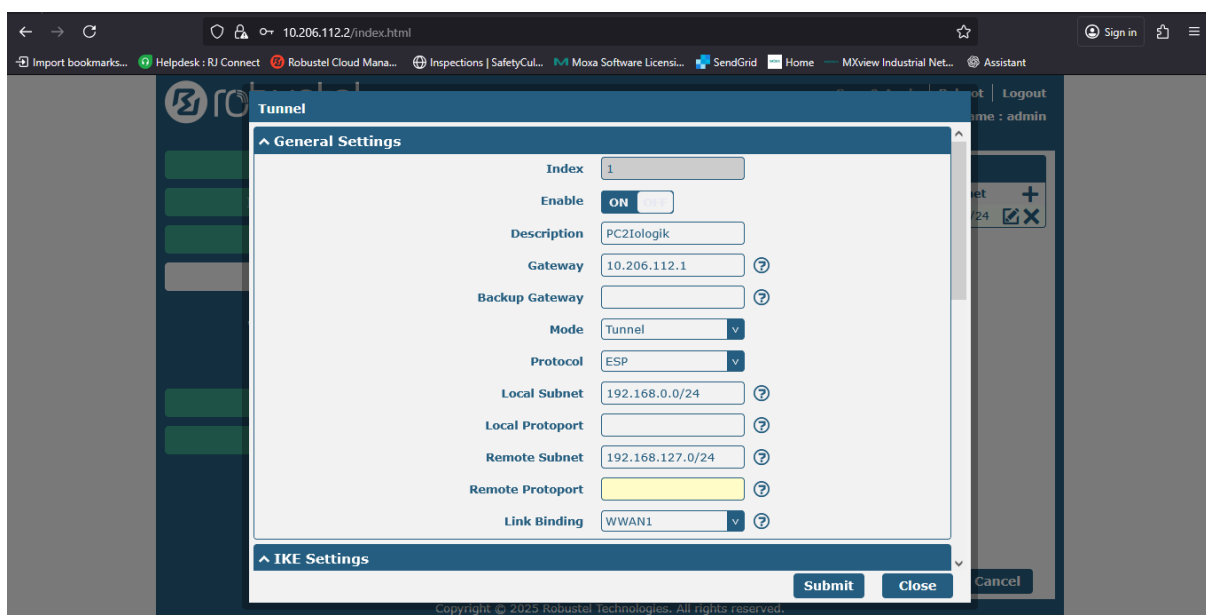
Configure the IPsec Tunnel

- Navigate to **VPN → IPsec → Tunnel** in the Robustel web interface.
- Click **Add** to create a new IPsec tunnel.
- Enter the required tunnel parameters according to your network design or VPN configuration plan.



Configure IPsec Settings

- After clicking **Add**, configure the **IPsec tunnel settings** according to your preferred or required VPN configuration.
- **Important:** All IPsec parameters (e.g., encryption, authentication, pre-shared key) **must match** between both Robustel routers, one on the SCADA side and the other on the PLC side, for the tunnel to establish successfully.



IPsec Settings continues:

← → ↺ 10.206.112.2/index.html

Import bookmarks... Helpdesk : RJ Connect Robustel Cloud Mana... Inspections | SafetyCul... Moxa Software Licens... SendGrid Home MXview Industrial Net... Assistant

Robustel Technologies

Logout
Username: admin

Tunnel

^ IKE Settings

IKE Type	IKEv1	▼
Negotiation Mode	Main	▼
Encryption Algorithm	3DES	▼
Authentication Algorithm	SHA1	▼
IKE DH Group	DHgroup2	▼
Authentication Type	PSK	▼
PSK Secret	●●●●●●	
Local ID Type	Default	▼
Remote ID Type	Default	▼
IKE Lifetime	86400	?

^ SA Settings

Encryption Algorithm	3DES	▼
Authentication Algorithm	SHA1	▼

Submit Close Cancel

IPsec Settings Continue:

10.206.112.2/index.html

Sign in

Assistent

Import bookmarks... Helpdesk : RJ Connect Robustel Cloud Mana... Inspections | SafetyCul... Moxa Software Licensi... SendGrid Home MXview Industrial Net...

Logout
name : admin

Tunnel

PSK Secret

Local ID Type

Remote ID Type

IKE Lifetime

Default

Default

86400

SA Settings

Encryption Algorithm

Authentication Algorithm

PFS Group

SA Lifetime

Enable DPD

DPD Interval

DPD Failures

DPD Action

3DES

SHA1

PFS(N/A)

28800

ON

30

150

restart

Advanced Settings

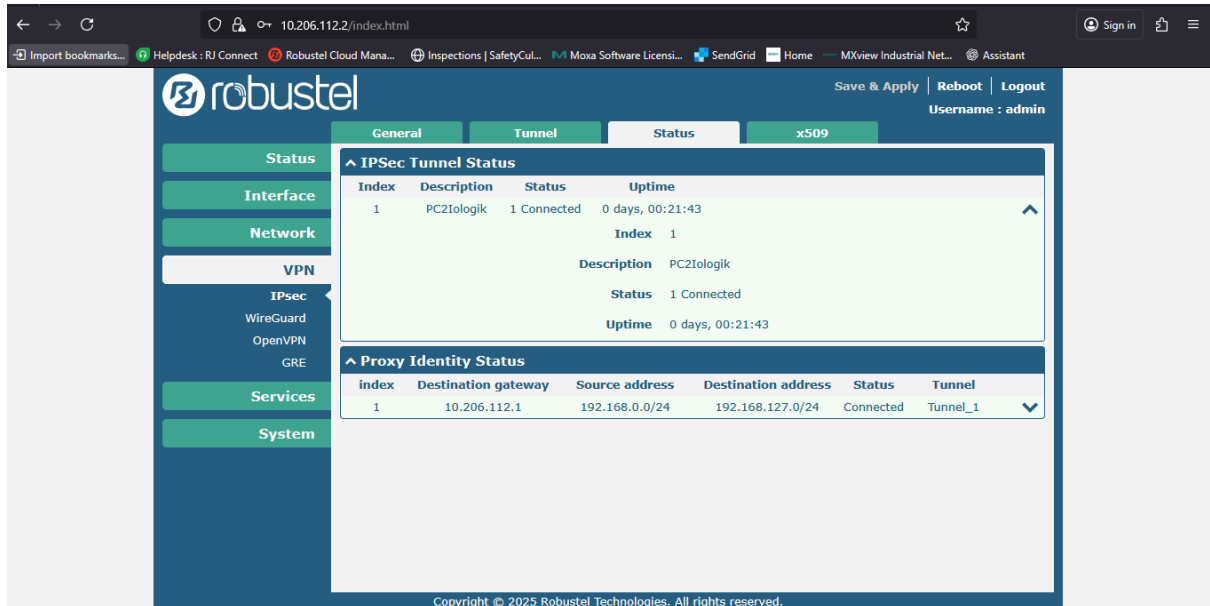
Submit

Close

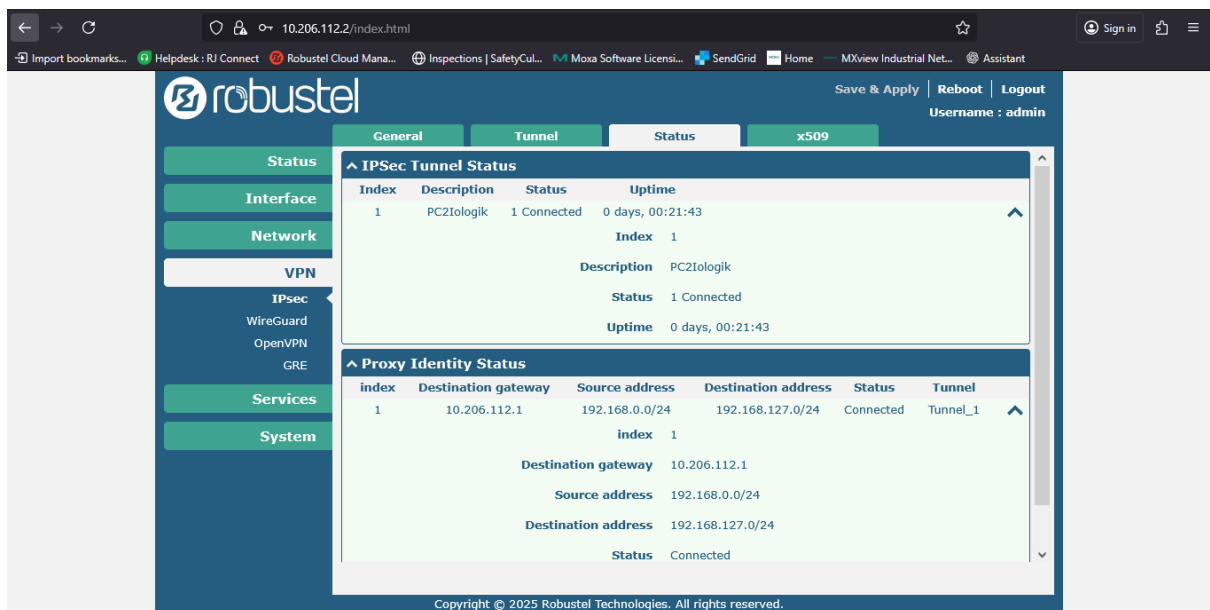
Cancel

Confirm VPN Connection

- Verify that the IPsec VPN tunnel shows as **Connected** in the Robustel web interface.
- This confirmation is only possible after the tunnel is properly configured and active on the other Robustel router (PLC side).



Continuation of status of VPN connection:

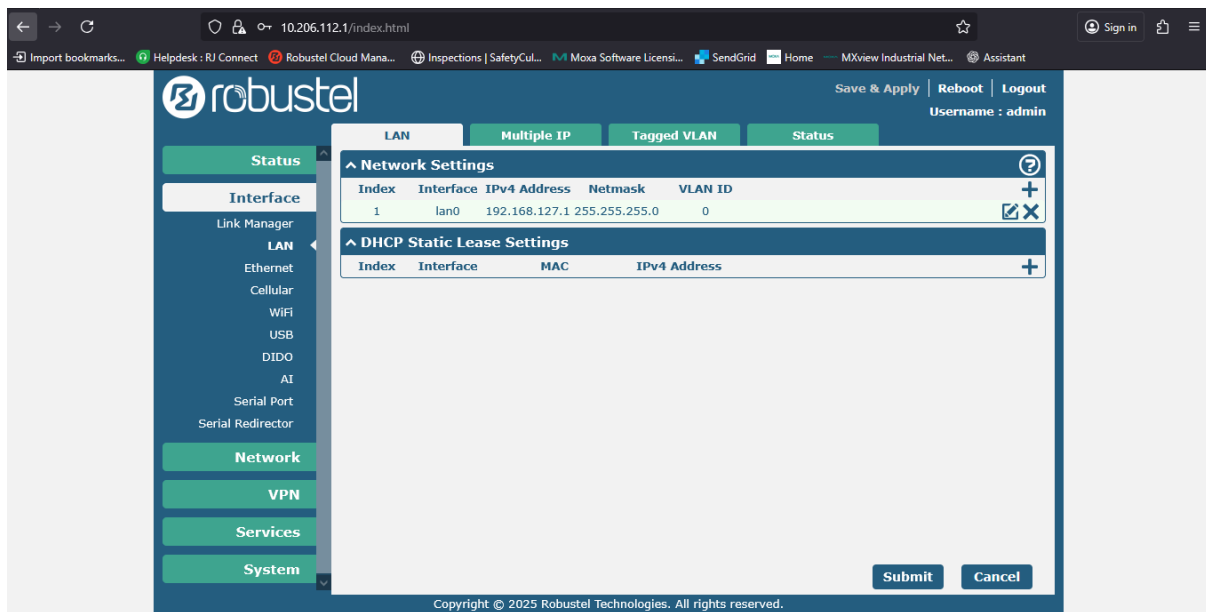


Configuration (Robustel PLC Side)

Note: Repeat the previous steps to configure the WWAN settings on the second Robustel router located on the PLC side. Ensure both routers have matching IPsec tunnel configurations for successful VPN connectivity.

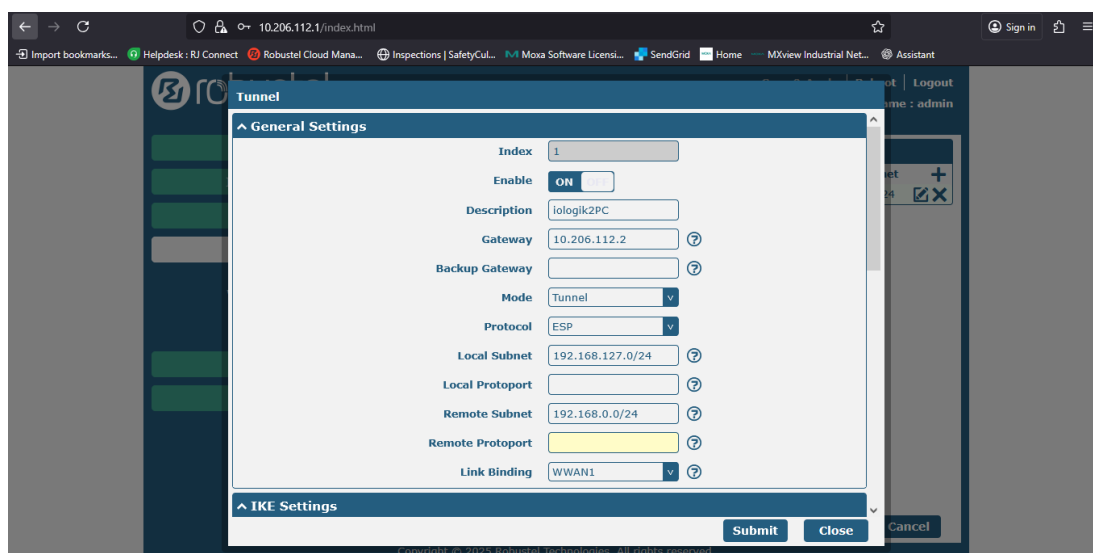
Configure LAN IP on PLC-Side Router

- Change the default LAN IP address of the router to match the subnet range of the PLC/Iologik network.
- After making changes, click **Submit**, then **Save & Apply** to apply the new settings.



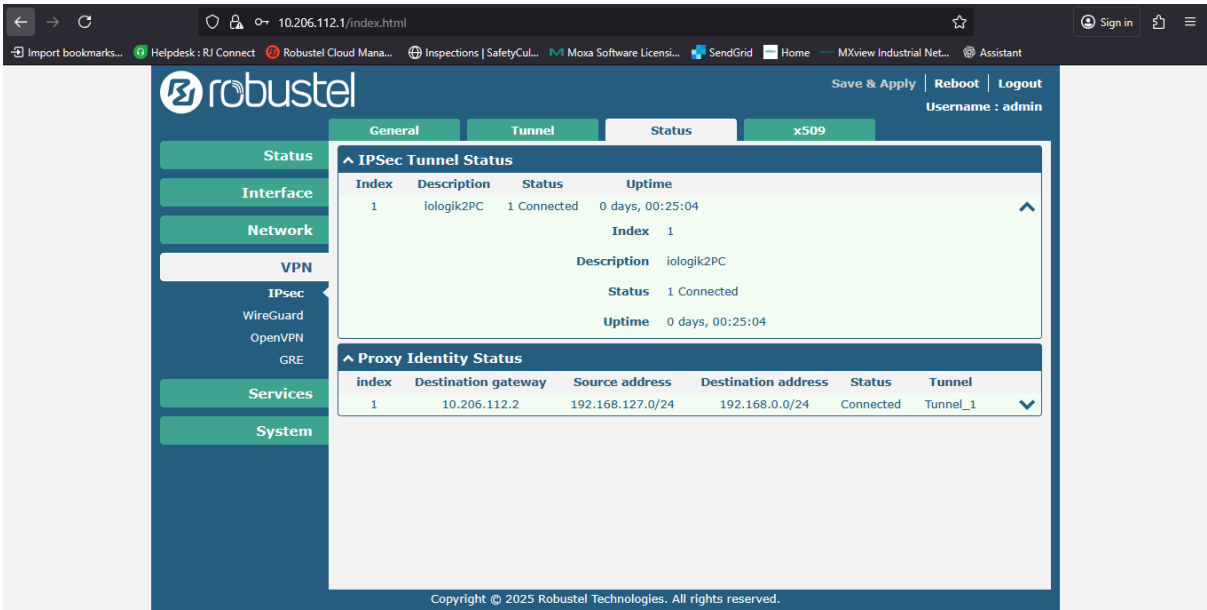
Configure IPsec VPN Tunnel (PLC-Side)

- Ensure all **IPsec VPN tunnel settings** match the configuration of the Robustel on the SCADA side.
- Confirm that the **pre-shared key** is **identical** on both routers; any mismatch will prevent the tunnel from connecting.

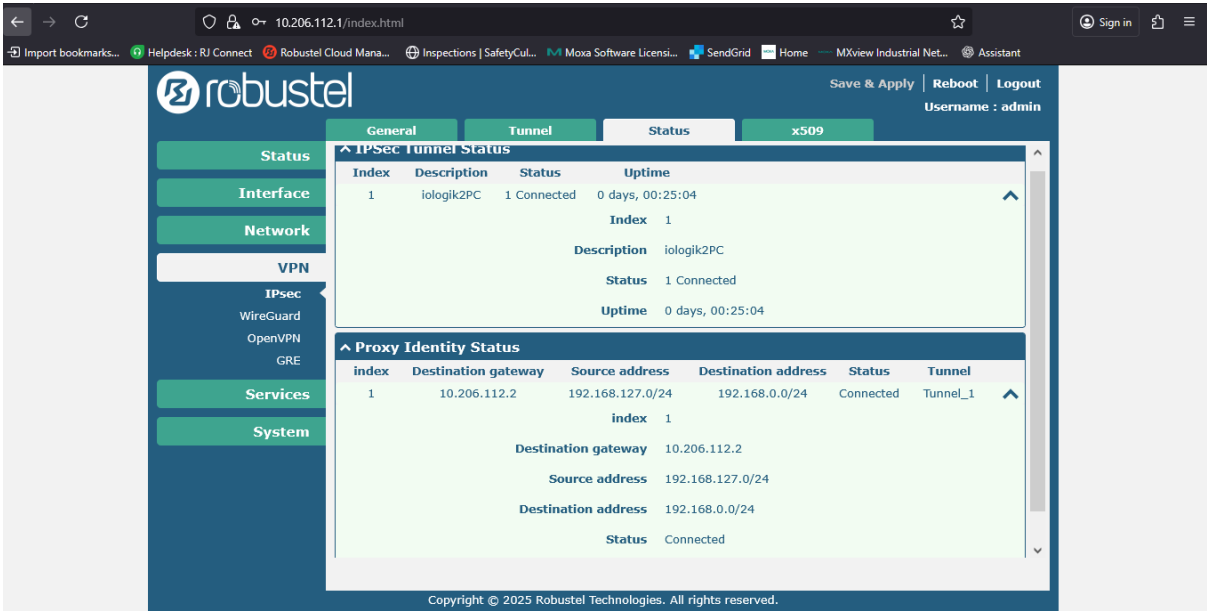


Verify VPN Tunnel Status

- Check that the **IPsec VPN tunnel** has successfully established a connection with the other Robustel.
- The **Status** should display “**Connected**” in the web interface.



Continuation of Tunnel status



Configuration (ioLogik E1242)

- Configure the **Ethernet settings** on the iologik device.
- Assign an **IP address** within the same subnet as the Robustel router's LAN IP (e.g., **192.168.127.x**).
- Set the **default gateway** to the Robustel router's LAN IP address (e.g., **192.168.127.1**).

The screenshot shows the Moxa ioLogik Remote Ethernet I/O Server web interface. The browser address bar shows <http://192.168.127.254/home.htm>. The page title is "ioLogik Remote Ethernet I/O Server" and the URL is www.moxa.com. The interface displays the following information:

Model	Name	Location	IP	Serial No.	System Elapsed Time	MAC Address	Firmware
E1242 Ethernet I/O Server			192.168.127.254	TAHGB1016330	00:23:16	00-90-e8-70-fe-43	V3.4 Build24072310

The "Ethernet Configuration" section is active, showing the "Ethernet Parameters" tab. The "IP Configuration" is set to "Static". The "IP Address" is 192.168.127.254, the "Subnet Mask" is 255.255.255.0, and the "Gateway" is 192.168.127.1. A "Submit" button is visible at the bottom of the configuration section.

Test Iologik Connectivity Using Modscan

- Open **Modscan** on your PC.
- Enter the **IP address** of the iologik device configured earlier.
- Click **OK** to establish the connection and begin communication.

The screenshot shows the ModScan32 - ModSca1 application window. The "Connection Details" dialog box is open, displaying the following configuration:

Connect Using: Remote modbusTCP Server
IP Address: 192.168.127.254
Service Port: 502

Configuration:

- Baud Rate: 115200
- Word Length: 8
- Parity: NONE
- Stop Bits: 1

Hardware Flow Control:

- ☐ Wait for DSR from slave
- ☐ Wait for CTS from slave

DTR Control: Disable
RTS Control: Disable

Delay: 0 ms after RTS before transmitting first character
Delay: 0 ms after last character before releasing RTS

Protocol Selections: OK Cancel

The background window shows the "ModSca1" interface with the "Address" field set to 0001 and the "Length" field set to 4. The "Device" field is set to 01: COIL ST. The status bar at the bottom indicates "Polls: 0" and "Resps: 0".

Confirm ioLogik Connection

- The iologik device should successfully connect, allowing you to monitor its status from your PC via Modscan.
- This connection is established securely through the **IPsec VPN tunnel** configured on the Robustel routers.